ROHIN LENGADE

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PROFESSIONAL SUMMARY

Solutions Engineer with 2+ years owning the full robotics automation customer lifecycle at GrayMatter Robotics — pre-sales technical analysis and ROI validation, proof-of-concept and workshop-driven process development, on-site deployment, and post-activation support and account expansion. M.S. in Aerospace Engineering (Advanced Design, Manufacturing & Robotics) from USC, with hands-on depth in robotic process development, system testing, and parametric CAD. Skilled at translating ambiguous customer requirements into de-risked, scalable automation solutions and comfortable operating across sales, engineering, and customer-success functions.

EXPERIENCE

GRAYMATTER ROBOTICS (GMR) — Feb 2024 – Present

Robotics Engineer II, Solutions Engineering Team

Pre-Sales & ROI Analysis

- Served as the lead technical resource on sales calls, fielding in-depth engineering questions and validating that proposed robotic solutions met customer ROI and payback targets before proposal sign-off.
- Presented proof-of-concept results and built customer-specific proposals, translating technical feasibility into a clear business case for automation investment.

Technical Discovery & Feasibility

- Partnered directly with customers to capture process requirements, throughput targets, cycle-time constraints, and existing factory layout, ensuring proposed cell designs matched real production conditions.
- Analyzed robot reach, achievable performance, and part coverage to de-risk projects ahead of contract commitment, surfacing layout and cycle-time issues before they became costly change orders.

Parametric CAD Tooling

- Designed and built a fully parametric CAD tool that rapidly configures robotic cell layouts and auto-extracts the key dimensions needed for deployment, plus the parameters for reach and coverage studies — replacing manual, one-off CAD work with a repeatable process across the Solutions Engineering team cutting turnaround time from 5 days to 8 hours.

Post-Activation Support & Account Expansion

- Owned post-activation support for live customer cells, providing remote and on-site troubleshooting and operator support to protect production uptime after handoff.
- Lead expansion engagements with existing accounts — adding new SKUs into already-deployed cells and scaling installations from single-cell to multi-cell lines — growing automation footprint within the existing customer base.

Robotics Engineer, Systems & Applications Team

Process & Application Development

- Lead process development for robotic grinding applications and deployed the systems on customer sites to process industrial battery storage units, fire truck and ambulance chassis, and aircraft engine lipskins.
- Led process development for single and dual robot sanding, paint-removal, and polishing applications, deploying systems for fire trucks, cement mixer drums, luxury cabinetry, wood/granite countertops, and aircraft interior cabinetry.

Hardware & Software Integration

- Developed and deployed an automated tool-changer enabling dynamic swapping between grinding and sanding processes within a single part cycle.

System Calibration

- Performed intrinsic/extrinsic camera (Zivid, LMI, Creaform, Leica), turntable, force-compliant device, and tool-center-point calibration at customer sites and in-house, achieving sub-1.5mm system tolerance.

Customer Relations & Workshops

- Organized and led customer workshops, weekly sync meetings, and pre-activation reviews to set and meet expectations for each robot cell before deployment.
- Collaborated with customers to integrate cell(s) into their production lines, delivering a cohesive automation system unique to each customer, then trained operators and provided remote assistance to sustain solution uptime.

USC CENTER FOR ADVANCED MANUFACTURING (CAM) — Dec 2022 – Feb 2024

Robotics Engineer, Mechanical Design & Integration

- Developed a modular pneumatic vacuum suction gripper (5g–9kg payload) for an articulated collaborative robot (Kuka IIWA-7), with Arduino–ROS interfacing and a two-robot RealSense-based experimental setup.
- Engineered a remote-operated precision screwdriver with RealSense vision and ultrasonic sensing for a 20% increase in depth-sensing accuracy, including a Mask-RCNN model for threaded-hole detection.
- Engineered a remote-operated high-pressure vacuum lid that improved automated composite debulking vacuum generation by 30%.

USC LIQUID PROPULSION LABORATORY (LPL) **Sept 2022 – Jan 2024**

Responsible Engineer, Propulsive Landing Hopper

- Led a team of 20 graduate students to engineer USC's first liquid-fueled propulsive landing hopper.
- Spearheaded Thrust Vector Control technology to gimbal a 675 lb. additively manufactured re-gen engine.
- Designed, manufactured, and tested mechanical systems for LPL's first Ker-Lox static-fire test stand, including a cooling/filtering system for the Data Acquisition box and an overhaul of cryogenic-rated high-pressure valves.

NASA JET PROPULSION LABORATORY (JPL) **Oct 2021 – Aug 2022**

Research Intern, 357H: Materials Development & Manufacturing Technology

- Implemented lattice optimization and thermo-structural analysis of thermal brackets with anisotropic material, increasing thermal & structural performance of an optical instrument support structure by 20%.
- Fabricated topology-optimized satellite structural components using Laser Powder Bed Fusion

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA **2022 – 2024**

Master of Science, Aerospace Engineering (emphasis: Advanced Design & Manufacturing in Robotics) GPA: 3.4

Directed Research: Aerodynamic analysis of camber-morphed airfoil in autonomous morphing of airfoil geometry

- Composites Design & Manufacturing: Manufactured, tested & analyzed aerospace-grade composite panels
- Additive Manufacturing: Optimized part geometry for 3D printing & manufacturing using FDM & SLA
- Robotic Mobility: Built a quadruped to traverse obstacle-induced terrain with minimal control system
- Spacecraft & Rocket Propulsion: Designed solid, liquid, electric & nuclear propulsion engines

KLS GOGTE INSTITUTE OF TECHNOLOGY **2018 – 2022**

Bachelor of Engineering, Mechanical Engineering GPA: 3.65

Final Thesis (with NASA JPL): Topology & Lattice Multi-Physics Optimization of next-generation spacecraft structural components

TECHNICAL SKILLS

Technical Discovery and Requirement Gathering • Proof-of-Concept Development • Proposal Development and Technical Presentation • Computer-Aided Design (CAD) • Python • Robot Operating System (ROS) • Topology & Lattice Optimization • Additive Manufacturing • Design, Manufacturing & Handling of Composites/Carbon Fiber • Machine Shop, & Power Tools

Equipment: EOS M290 metal 3D printer, Stratasys F120, Haas CNC, Formlabs Form 3+

PUBLICATIONS

- Omey M., Santosh N., Rohin L., “Physics-Informed Learning to Enable Robotic-Screwdriving Under Hole Pose Uncertainties,” IEEE International Conference on Intelligent Robots & Systems (IROS), 2023.
- Rohin L., “A Prototype Aerospike: Another Fish in the Sea,” ASME International Mechanical Engineering Congress & Exposition (IMECE), Vol. 4: Advances in Aerospace Technology, 2020.

HONORS & AWARDS

- Demonstrated Thrust Vector Control of 675 lb. liquid bi-propellant engine, making USC LPL one of two collegiate teams in the world to thrust-vector a rocket engine (Dec 2023); featured in the IAF newsletter (Feb 2024).
- Smart Robot Assistant for Safe Tool Hand-Over featured in IEEE Spectrum (March 2023).
- Awarded Best Student in Mechanical Engineering upon receiving Bachelor's degree (April 2022).

AFFILIATIONS

- USC Center for Advanced Manufacturing (USCCAM), Robotics Engineer
- USC Liquid Propulsion Laboratory (USCLPL), Responsible Engineer
- American Society of Mechanical Engineers (ASME), Member & past Lead at KLSGIT chapter